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Cloud Computing & Security [BIS613D] Report on

“CLOUD BASED FILE SHARING SYSTEM”

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CLOUD- BASED FILE SHARING SYSTEM

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1. Introduction

A Cloud-Based File Sharing System is a secure, scalable platform that allows users to store, manage, and distribute digital files over the internet. Traditional file-sharing methods, like physical drives or email attachments, often suffer from storage limits, security risks, and versioning issues.

This project implements a centralized cloud directory where users can access their data from any device with an internet connection. It provides a modern alternative to traditional FTP servers, offering synchronized access and real-time collaboration features.

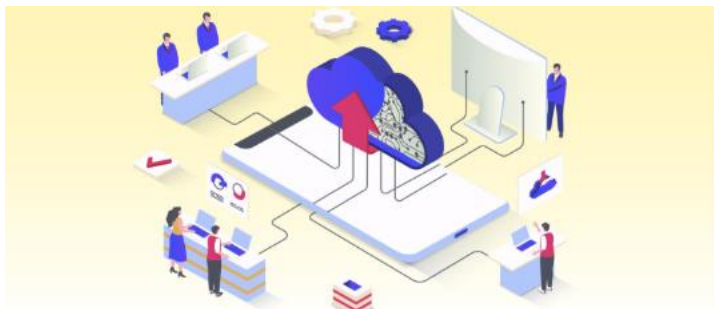
2. Objectives

- To develop a secure cloud platform for remote file storage.
 - To implement multi-user access with granular permissions (view, edit, share).
 - To ensure data integrity through encryption and hashing to prevent duplicate uploads.
 - To provide a user-friendly interface for seamless file management.
 - To enable real-time file synchronization across multiple devices.
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3. System Architecture

3.1 Architecture Overview

The system uses a cloud-based architecture where all components interact through Firebase services. The frontend communicates directly with the cloud backend, ensuring real-time updates and scalability.



3.2 Frontend (Client Side)

The frontend is developed using React.js and provides an interactive interface for users.

Functions include:

- User login and registration
- Upload and download files
- View shared files
- Manage file access

3.3 Backend (Firebase Services)

Firebase acts as the backend and handles all server-side operations.

Functions:

- User authentication
- File storage handling
- Database operations
- Secure file access

3.4 Database

The database (Firestore or Realtime Database) stores:

- User details
- File information
- File permissions
- Sharing links

It ensures that all users see updated data instantly.

3.5 Authentication Module

- Secure login/signup
 - User session management
 - Data protection
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3.6 Admin Module

The admin panel provides control over the system.

Features:

- Monitor users
 - Manage files
 - Control permissions
 - View system activity
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4. Working Principle

The system works based on real-time interaction between users and the cloud database.

Steps:

1. User logs into the system
2. Uploads files to cloud storage
3. Files are stored in database
4. User can share files via link or permission
5. Other users can access/download files
6. Admin monitors system



5. Key Features

- Secure file upload/download
 - Cloud storage access
 - File sharing via links
 - User authentication
 - Real-time access
 - Admin dashboard
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6. Advantages

- Access files anywhere
 - Secure data storage
 - Easy sharing
 - Scalable system
 - Time-saving
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7. Conclusion

The Cloud-Based File Sharing System provides a secure and efficient way to store and share files using cloud technology. It improves accessibility, enhances security, and demonstrates practical implementation of cloud computing.